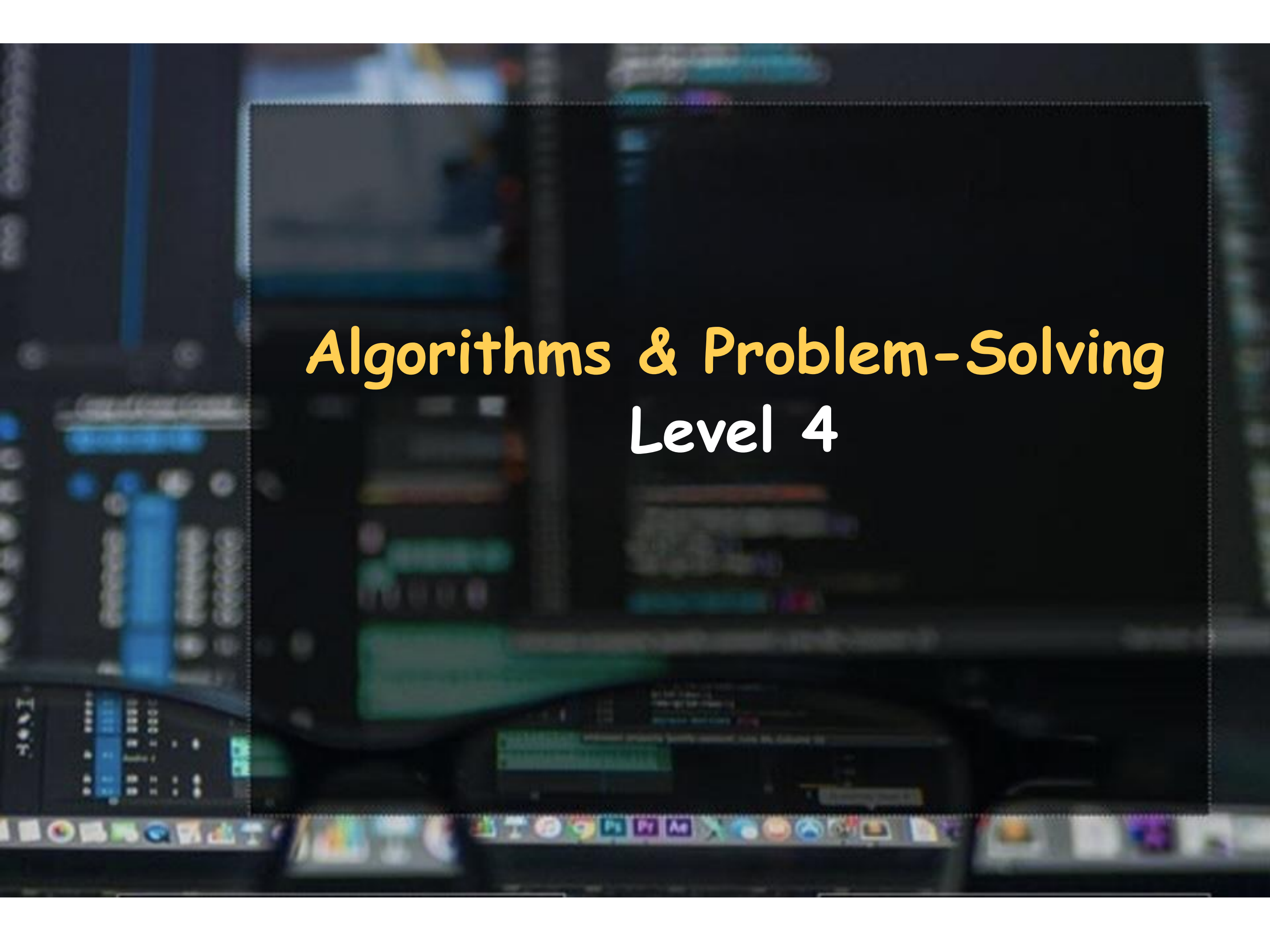
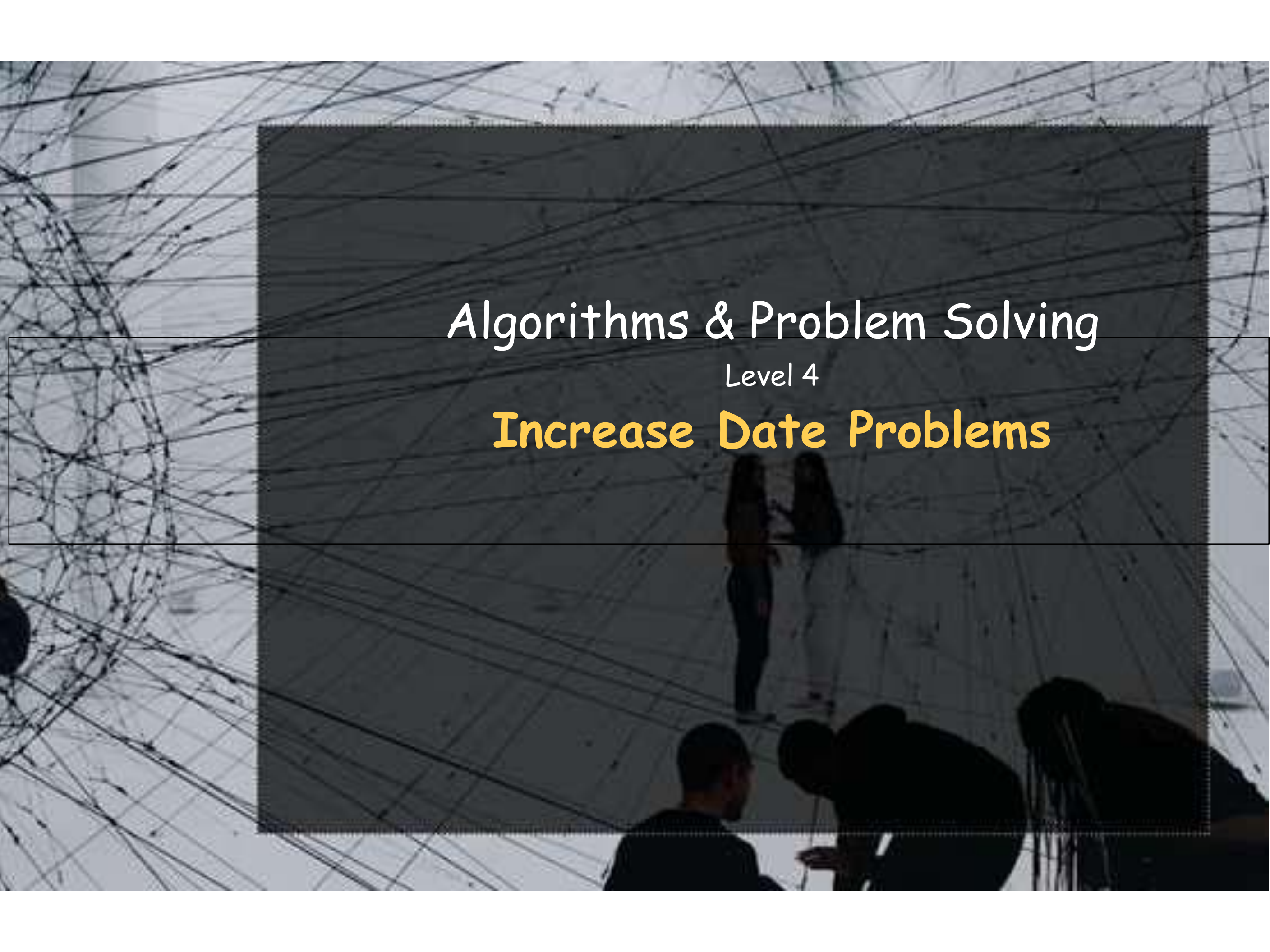




Algorithms & Problem-Solving Level 4



Algorithms & Problem Solving

Level 4

Increase Date Problems



Definition

- The program's objective is to read a date from the user and then increment it by a specified number of days, weeks, months, years, decades, centuries, or millennia using separate functions for each type of increment.

Procedures

- 1 Defining the sDate structure:
 - It contains three values: `Year`, `Month`, `Day`
- 2 Reading inputs:
 - Via functions:
 - ReadDay()
 - ReadMonth()
 - ReadYear()
 - and combining them into ReadFullDate()
- 3 Checking for leap years:
 - The `IsLeapYear(year)` function determines whether February has 29 days.
- 4 Number of days in each month:
 - NumberOfDaysInAMonth(month, year)` returns the correct number based on the month and year.
- 5 Increasing by one day:
 - IncreaseDateByOneDay(date)` increments the date by one day, taking into account the transition to the new month or year.
- 6 Increment by a specific number of days:
 - IncreaseDateByXDays(days, date) Calls the previous function the required number of times

7 Increment by a week or several weeks:

- IncreaseDateByOneWeek() Increments by 7 days

- IncreaseDateByXWeeks(weeks, date) Should increase by 7 x weeks (but the current code only increases by one day per week !)

- 📌 Note: There is a minor logic error: The number of weeks should be multiplied by 7 inside the loop

8 Increasing a month or multiple months:

- IncreaseDateByOneMonth() adds one month, ensuring the day doesn't exceed the maximum limit for the new month.

- IncreaseDateByXMonths(months, date) calls the previous one multiple times.

9 Increasing a year or more:

- IncreaseDateByOneYear() adds a year, adjusting February 29th to the 28th if needed.

- IncreaseDateByXYears() repeats the process.

- IncreaseDateByXYearsFaster() adds years directly in one step.

- Decades:

 - IncreaseDateByOneDecade() adds 10 years.

 - IncreaseDateByXDecades() adds (decade x 10) years slowly. (Gradually)

 - IncreaseDateByXDecadesFaster() adds it directly in one step

- Century:

 - IncreaseDateByOneCentury() adds 100 years

- Millennium:

 - IncreaseDateByOneMillennium() adds 1000 years

Example:

Please enter a Day? 31

Please enter a Month? 12

Please enter a Year? 2022

Date After:

01-Adding one day	is: 1/1/2023
02-Adding 10 days	is: 11/1/2023
03-Adding one week	is: 18/1/2023
04-Adding 10 weeks	is: 29/3/2023
05-Adding one month	is: 29/4/2023
06-Adding 5 months	is: 29/9/2023
07-Adding one year	is: 29/9/2024
08-Adding 10 Years	is: 29/9/2034
09-Adding 10 Years (faster)	is: 29/9/2044
10-Adding one Decade	is: 29/9/2054
11-Adding 10 Decades	is: 29/9/2154
12-Adding 10 Decade (faster)	is: 29/9/2254
13-Adding One Century	is: 29/9/2354
14-Adding One Millennium	is: 29/9/3354

Summary

- ✓ The program demonstrates the power of structured programming using functions.
- ✓ It can be easily expanded to calculate other time periods or different calendars.

Thank you 😊